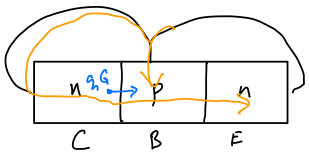
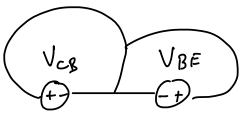
ECE 4570

Q 6.14



$\Leftrightarrow$



a) Let I_{Bo} be equilibrium base current
I_{Co} be equilibrium collector current
I_{Eo} be equilibrium emitter current

Note: $I_{Co} = \alpha_F I_{Eo}$.

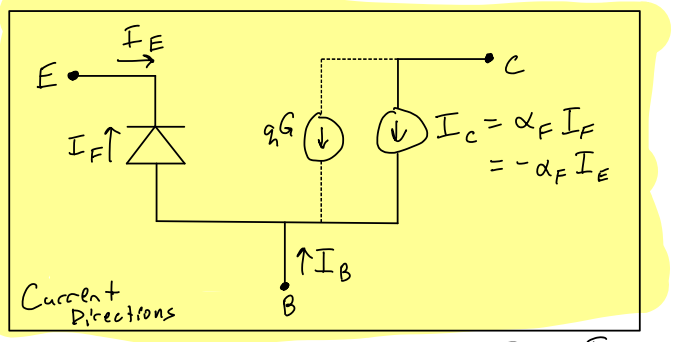
$\Rightarrow q_nG$ added to $I_{co} \Rightarrow$ balanced by reduction in base current.

$\Rightarrow I_C = I_{Co} + q_nG, \quad I_B = I_{Bo} - q_nG$

I_{Eo} should be same; doesn't see diff (in reverse direction)

$$\Rightarrow \boxed{\begin{aligned} I_E &= -I_{Eo} \\ I_B &= I_{Bo} - q_nG \\ I_C &= I_{Co} + q_nG \end{aligned}}$$

b) Current directions defined to right $\rightarrow$



$\Rightarrow I_B$ is constant but voltage may change. $\Rightarrow I_B = I_{Bo}$

I_C is still $q_nG + \alpha_F I_F$, $|I_{Eo}| = |I_F|$

$$I_F = I_B + I_{Co} + q_nG = I_B + \alpha_F I_F + q_nG \Rightarrow I_F = \frac{I_B + q_nG}{1-\alpha_F} = \frac{\beta_F}{\alpha_F}(I_B + q_nG)$$

Note $I_E = -I_F$

$$I_C = I_F \alpha_F + q_nG = \beta_F(q_nG + I_B) + q_nG$$

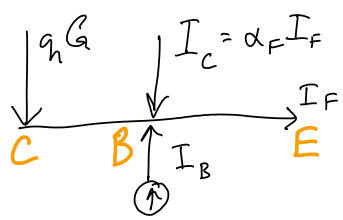
Collecting, $\boxed{I_B = I_{Bo}}$

Current Directions defined above

$$\boxed{I_C = \beta_F(q_nG + I_B) + q_nG}$$

$$\boxed{I_E = -\frac{I_B + q_nG}{1-\alpha_F}}$$

Diagram



Q 7.2

- The paper basically comments that interest in heterogeneous BJTs have increased recently.
- It then states that it aims to generalize two formulas for BJTs so that they can also be used for heterogeneous BJTs.
- Specifically it mentions that it will generalize the formulas for tau (the transit time for an electron to cross the base) and Jn (the current flow density in the base)

- For tau, after some derivation, it arrives at the more general formula in equation 12
  - tau= integral_0^w [ni^2/p * integral_z^w [p/(dn*ni^2)] dy] dz
- For Jn, after some derivation, it arrives at the more general formula in equation 10
  - Jn= -q* exp(qVBE/kT) / (integral_B[p/(dn*ni^2)]dx)

It then concludes with an example of using these formulas.
This example is of a linearly graded HBT (like below) and derives some useful relations.

Q 7.3

a) Depletion Widths are $\sim 1\,nm \Rightarrow$ basically zero.

$W_B = 200nm$

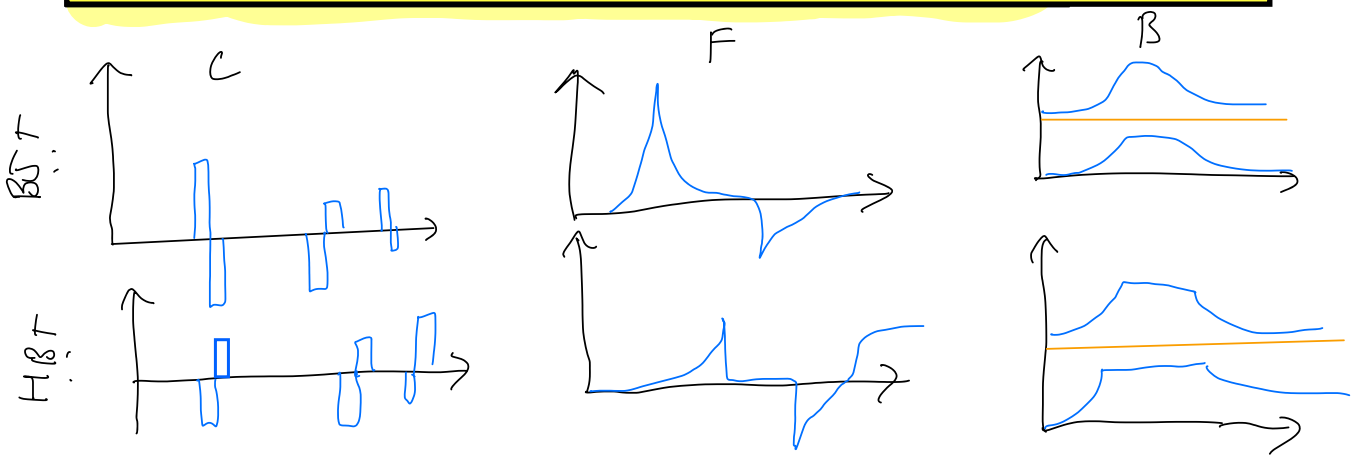
$E_g(GaAs) = 1.424eV$ $\quad E_g(Al_{0.3}Ga_{0.7}As) = 1.424 + 1.247 \cdot 0.3 = 1.798eV$

$E_g(Al_{0.1}Ga_{0.9}As) = 1.424 + 1.247 \cdot 0.1 = 1.5487\ eV$

In Base, $E_g = 1.5487 - \frac{1.5487 - 1.424}{200nm} x$ if $x=0$ is E-B interface.

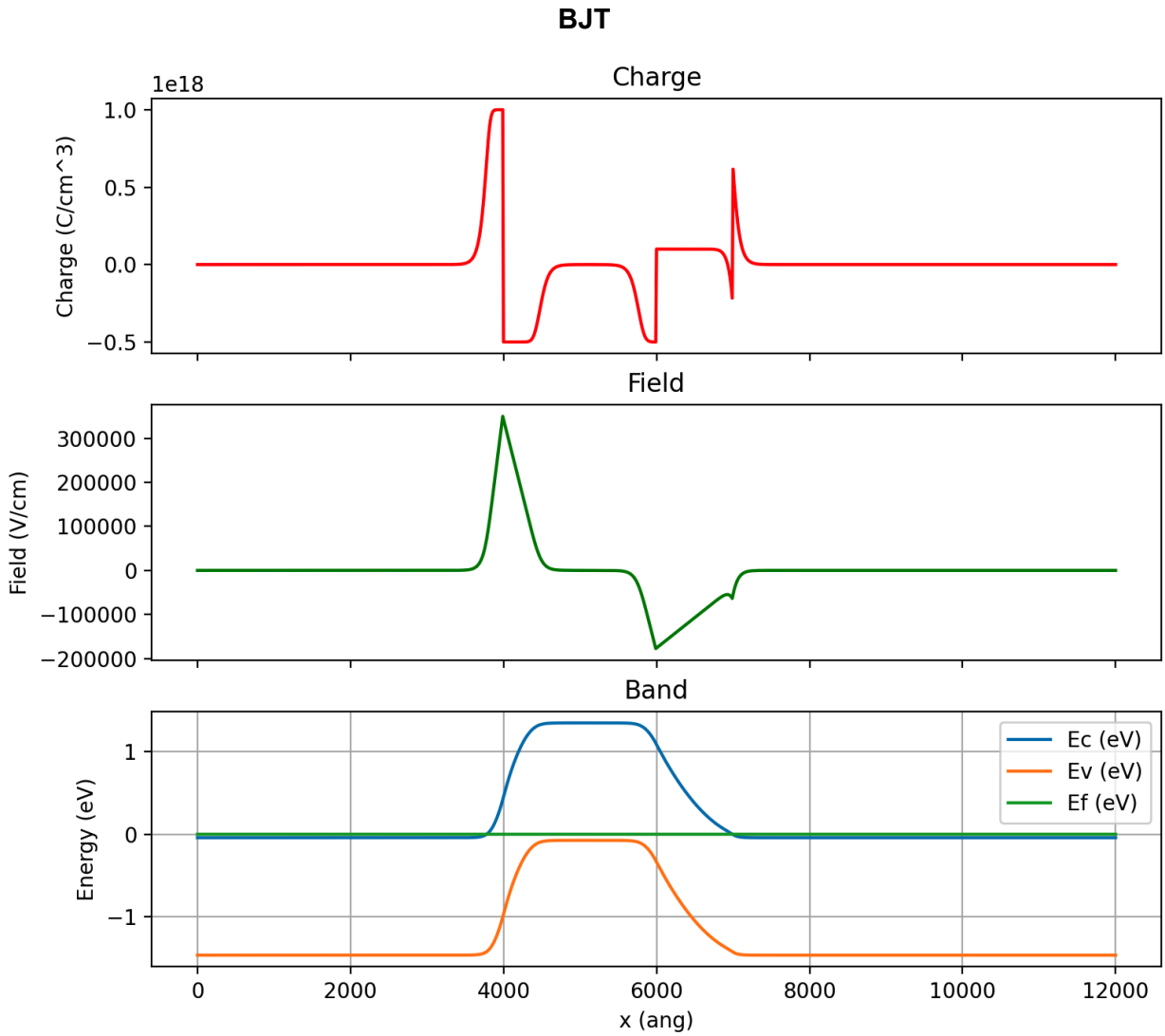
$\Rightarrow E_B = \frac{\partial E_g}{\partial x} = \boxed{E_B \approx -6235 \frac{V}{cm}}$ Simulation says $\sim -3000 \frac{V}{cm}$ ✓ Reasonable

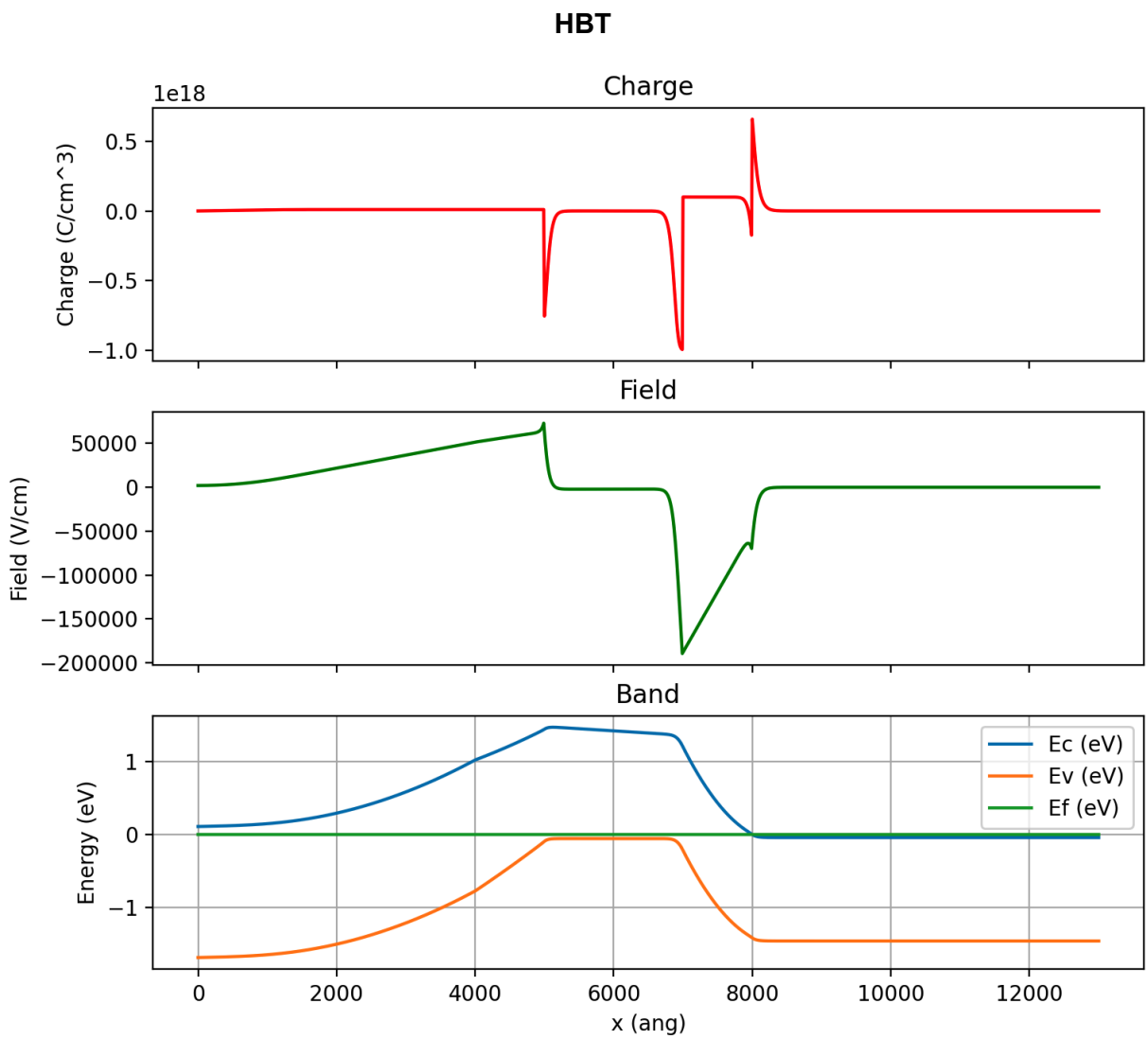
The sub-collector's purpose is to minimize the collector's resistance. Intuitively, it ensures a smooth transition out from the transistor and enhances current stability.



Q3

1D Poisson Plots for Q3




HBT
Charge
1e18
Charge (C/cm^3)
0.5
0.0
−0.5
−1.0
Field
Field (V/cm)
50000
0
−50000
−100000
−150000
−200000
Band
Energy (eV)
1
0
−1
Ec (eV)
Ev (eV)
Ef (eV)
0
2000
4000
6000
8000
10000
12000
x (ang)

b) $J_c = -q D_n \frac{dn_B}{dx} + q\mu_n n_B E_B$

$D_n = 200, \quad D_p = 10$

$\mu_n = 8500 \frac{cm^2}{V \cdot s}$

$$\frac{dn_B}{dx} = \frac{q\mu_n n_B E_B - J_c}{q D_n}$$

$\Rightarrow$ Solving using integrating factors, $n_B = \frac{J_c}{q\mu_n E_B} + C_1 \exp\left(\frac{\mu_n E_B}{D_n} x\right)$

BC: At $x = W_B$, $n_B(W_B) = 0$

$$\Rightarrow \quad C_1 \exp\left(\frac{\mu_n E_B W_B}{D_n}\right) = -\frac{J_c}{q\mu_n E_B} \Rightarrow C_1 = -\frac{J_c e^{-\frac{\mu_n E_B W_B}{D_n}}}{q\mu_n E_B}$$

$$\Rightarrow \quad \boxed{n_B = \frac{J_c}{q\mu_n E_B}\left[1 - \exp\left(\frac{\mu_n E_B}{D_n}(x - W_B)\right)\right]}$$

μ_n, D_n defined above

c) For HBT

From Kroemer,

$$\tau = \int_0^{W_B} \frac{n_i^2}{p}\left(\int_z^{W_B} \frac{p}{D_n n_i^2} dy\right) dz, \quad \text{but} \quad n_B = \frac{n_i^2}{p}$$

From his example, using appropriate approximations we see: $\tau = \frac{W_B}{\mu_n E_B}\left(1 - \frac{kT}{q E_B W_B}\left(1 - \left(\frac{n_B(0)}{n_B(W_B)}\right)^2\right)\right)$

Which Kroemer approximates as (HBT)

$$\boxed{\tau_{tr} = \frac{W_B}{\mu_n E_B}\left(1 - \frac{kT}{q_n E_B W_B}\right)}$$

for normal operation

Subbing in values,

$$\boxed{\tau_{tr} = 2.99 \cdot 10^{-13}\ s}$$

(0.3ps) HBT

For BJT,

Kroemer gives: (BJT)

$$\boxed{\tau_{tr} = \frac{W_B^2}{2D_n} = 10^{-12}\ s}$$

BJT

(1ps)

Ratio:

$$\boxed{\frac{\tau_{tr}^{BJT}}{\tau_{tr}^{HBT}} = 3.34}$$

d) $\gamma_E = \dfrac{J_{nE}}{J_{nE} + J_{pE}}$

$$\gamma_E = \left(1 + \frac{G N_B D_p}{G N_E D_n}\right)^{-1}$$

(OK approx from TA)

For BJT,

$$G_{nB} = \frac{Q_{nBO}}{q} \approx N_{aB} \cdot W_B$$

$$G_{nE} \approx N_{dE} \cdot W_E \Rightarrow$$

$$\boxed{\begin{array}{ll} \gamma_E = 0.976 & \text{for BJT} \\ \alpha_T = 0.990025 & \end{array}}$$

$$\alpha_T = 1 - \frac{W_B^2}{2D_n \tau_{tr}} \Rightarrow$$

For HBT,

$$\gamma_E = \left(1 + \frac{G_1 N_B D_p}{G N_E D_n} e^{-\frac{\Delta E_g}{kT}}\right)^{-1} \Rightarrow$$

$$\boxed{\gamma_E = 0.9999 \text{ for HBT} \quad \alpha_T = 0.999985}$$

$$\alpha_T = 1 - \frac{W_B^2}{2 D_n \tau_{br}}$$

$$\beta_F = \frac{\gamma \alpha_T}{1 - \gamma \alpha_T}$$

$$\boxed{\beta_F = 29 \text{ for BJT}}$$

$V_{BE} = 0.8V$ means NO high injection or subthreshold effects on beta

$$\boxed{\beta_F = 8695 \text{ for HBT}}$$

HBT is dramatically better
Improvement in order of 100x higher

Part e/f on following page

Q3e)

Pro/Con Table

BJT	HBT
- Slower switching time - Higher transit time (slower) +Less complicated of a structure -Simpler and perhaps very slightly cheaper to produce -Lower quality factors/efficiency	**+ Importantly has a faster switching time (can switch at higher frequency)** + Lower transit time (faster) + Higher efficiency and gain (better quality factors) - More complicated of a structure + More complicated to analyze

Q3f)
Major players - Intel, IBM, and STMicroelectronics
Applications - automotive radar systems, advanced analog circuits, and satellite communications
Paragraph - After looking online and perusing a few papers and articles, I've noticed that Silicon-germanium heterojunction bipolar transistors are starting to gain in popularity since they offer a lot of useful capabilities. In particular, they are used for devices in and integration with BiCMOS technology. This combination offers significant advantages for high-speed applications, especially in telecommunications, such as 5G and future 6G networks. SiGe HBTs enable efficient performance in power amplifiers and low-noise amplifiers, providing a cost-effective solution without sacrificing performance. The ability of SiGe HBTs to operate across a broad frequency spectrum enhances their versatility, making them essential components in modern electronic systems and paving the way for future innovations in high-frequency technologies. These are all resultant from HBT ideas about mixing existing technologies to create a more robust transistor that can accommodate a wider array of frequencies. For instance, SiGe-based HBTs draw from traditional HBT concepts by incorporating germanium into the silicon structure. This addition enhances carrier mobility, allowing these transistors to operate at higher frequencies. The result is a technology that significantly boosts performance in high-speed applications.

Q 7.9

Charge Neutrality: $p = N_a + n \Rightarrow \int_0^{x_B} p(x)\,dx = \int_0^{x_B} N_a + n(x)\,dx$

Find $n(x)$:

$$n(0) = \frac{N_a}{2}\left[\left(1 + \frac{4n_i^2 e^{qV_{BE}/kT}}{N_a^2}\right)^{1/2} - 1\right], \quad n(x_B) = 0, \quad n'(x_B) = -\frac{n_i^2}{N_a}$$

$$\Rightarrow n(x) \approx n(0)\left(1 - \frac{x}{x_B}\right)$$

Find $p(x)$

$$p = N_a x_B + \tfrac{1}{2} n(0) x_B$$

And I_c:

$$I_c = \frac{q D_n n_i^2 A_E e^{qV_{BE}/kT}}{\int_0^{x_B} p(x)\,dx}$$

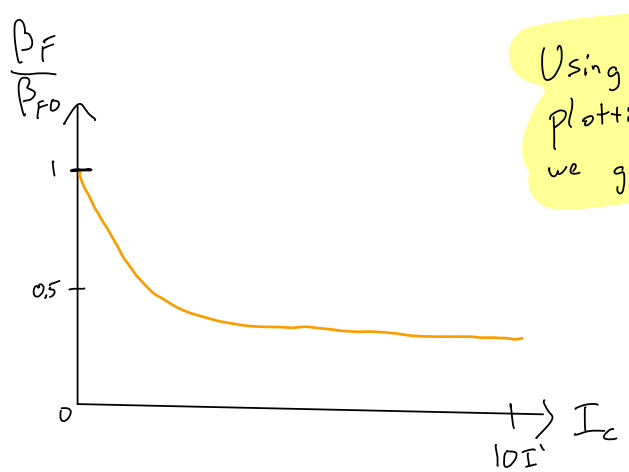
Now find $\frac{\beta_F}{\beta_{F0}} = \frac{I_c}{I_{c0}} = \frac{N_A x_B}{N_a x_B + \frac{1}{2} n(0) x_B} = \frac{1}{1 + \frac{n(0)}{2N_a}}$

Expand $n(0)$:

$$\frac{\beta_F}{\beta_{F0}} = \frac{1}{1 + \frac{1}{4}\left[\left(1 + \frac{4n_i^2 e^{qV_{BE}/kT}}{N_a^2}\right)^{1/2} - 1\right]}$$

Equivalently,

$$\boxed{\frac{\beta_F}{\beta_{F0}} = \left[\frac{3}{4} + \frac{1}{4}\left[1 + \frac{4 I_c x_B}{q D_n A_E N_a}\right]^{1/2}\right]^{-1}}$$



Using provided values, $\frac{q D_n A_E N_a}{x_B} = 16\text{mA}$ plotting with a graphing calc, we get the graph to the left